CS 131 SECOND EXAM

SOLVE FOR A TOTAL OF 50 POINTS. SHOW COMPLETE SOLUTIONS.

1. FIXEDPOINT THEOREM (/10)
   (a) Give the theorems for existence and uniqueness of a fixedpoint of a function *g(x)* on an interval and the convergence of the fixedpoint iteration. Illutrate this graphically. How do you use this to solve for the root of a function *f(x)*. Illustrate this with $f(x)=x^3-x+1$.
   (b) Derive Newton-Raphson iteration **for simple roots** starting from the fixed point iteration. What is the order of convergence of this method and why? Give the general form of the method for roots with multiplicity *k* that retain the order of convergence.

2. SOLVE ***ONE*** ITERATION OF THE FOLLOWING: (5/)
   (a) Solution of $\begin{matrix} 3x-2y-5z=1 \\ x^4-y^4+z^4=1 \\ xz-y^2=1 \end{matrix}$ using Seidel's Fixed Point Iteration with $\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
   (b) Minimum of $z=\left(y^2-x\right)^2+\left(x+1\right)^2$ using Newton's Method with $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
   (c) Nonlinear least squares solution for fitting $y=ax^b$ on the data set (2, 0.43), (3, 0.67), (5, 1.17) using Gauss-Newton's method with $\begin{bmatrix} a_0 \\ b_0 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

3. ILLUSTRATE GRAPHICALLY (2.5/)
   1. Regula Falsi Method
   2. Newton-Raphson Method
   3. Steepest Descent
   4. Nelder Mead

4. ONE-LINERS (1/)
   1. What theorem/s give/s the sufficient conditions for bisection method to work?
   2. When does regula-falsi method works no better than the bisection method?
   3. What is the order of convergence of the secant method?
   4. The drawback of iteration methods with higher order of convergence is ____________________________.
   5. How many function evaluations per iteration are made in golden section search?
   6. Redefine the constrained optimization problem $\underset{\forall x}{\operatorname{argmin}} f(x)$ subject to $g_1(x)=0$ and $g_2(x)=0$ using the method of Lagrange multipliers.
   7. Redefine the above problem by using static exterior penalty.
   8. State or give in equation form the Armijo's condition for the linesearch method of steepest descent method.
   9. Give the stopping condition for Nelder-Mead search algorithm without using the gradient.
   10. One iteration of Gauss-Newton method for solving nonlinear curve fitting is equivalent to _______________________________________.

5. What method/s from the second exam coverage that you like the most. Why? (5)